A metaplectic Jacquet–Langlands transfer for division algebras of prime degree, IV: Wild local correspondence and global trace formula comparison

Draft 5

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November 7, 2026

Abstract

We continue Parts I–III for an odd-degree covering of the multiplicative inner form of GL_r with r prime and $(r, n) = 1$. First we reconcile the split-diagonal cocycle with the ambient-cover restriction and central relabeling in Part I. The regular good-element criterion for *every* maximal torus, including nonsplit tori and wild residue characteristic, is already a theorem of Flicker–Kazhdan; we supply its proof with the parameters used in this series. We prove two refinements useful in global comparison: an almost-everywhere local-to-global criterion for globally good regular classes, and an exact local calculation of the kernel, index, and Haar modulus of the nth-power map on an arbitrary finite *étale* torus modulo scalars. The power-map calculation gives the split-place root multiplicities, while a separate regularity example explains why a regular root need not have a regular nth power. The genuine character transform of Part III is then placed alongside the corrected metaplectic character relation of Zou and the older work of Flicker–Kazhdan. A detailed wild rank-two example proves an integral, positive Fourier decomposition of the discriminant ratio on every quadratic maximal torus of the quaternion division algebra over $\mathbb{Q}_3(\zeta_3)$, including both ramified and unramified extensions; this is a torus-restriction result and does not prove irreducible-character integrality for the full group. A general toral Fourier criterion isolates the remaining nonabelian gluing question. A normalization ledger and explicit Flicker–Kazhdan-compatible split-place test-function conditions separate the previously established covered-inner-form trace comparison from the additional proposed comparison with ordinary $D^\times$. A conditional geometric comparison is made explicit with its global normalization factor, a necessary nongood-class support restriction, and separate split-place matching conditions. For $r = 2$, an unramified spherical-unit case follows from Flicker–Kazhdan in its standard normalization, subject to transport to the specified ambient cover. The wild irreducible-character integrality and the full global comparison are not established here.

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1 Introduction and mathematical status

The problem concerns three different local representation theories. Write $H_v = D_v^\times$, and let $\overline{G_{D_v}}$ denote its specified n-fold central cover; the cover $\overline{G}_v$ of $\mathrm{GL}_r(K_v)$ is defined inside the same ambient cover. An additional *ordinary* group H_v enters through the transform $L_{D,v}$ of Part III. These are not interchangeable. In particular, a comparison between $\overline{G_D}$ and $\overline{G}$ has already been obtained in Parts I–II (here cited as Draft 8 of each) under their stated test-function and spectral hypotheses. We cite Part III as Draft 9 throughout. The new proposed comparison between H and $\overline{G_D}$ is a different assertion.

Global assumptions. Let K be a totally imaginary number field with $\mu_n \subset K$, $n \geq 1$ odd. Let $r \geq 2$ be prime, $(r, n) = 1$, let L/K be cyclic of degree r, and let D be a central division K-algebra of degree r containing L. Put $S = \{v : D_v \text{ is division}\}$. Fix integers r', c such that

$$rr' \equiv 1 \pmod{n}, \qquad r - 1 + 2c \equiv 0 \pmod{n}, \qquad c_v \equiv r'c \pmod{n}. \tag{1}$$

Thus $r - 1 + 2rc_v \equiv 0 \pmod{n}$. In fact c_v is independent of v with these choices; the subscript recalls Part I's notation. Locally $F = K_v$ is a nonarchimedean field containing μ_n; p denotes its residue characteristic. We call v *wild* if $p \mid n$.

Status convention. **[P]** denotes a proof included here (possibly invoking **[E]** inputs established in cited publications or earlier parts); **[O]** denotes a proof outlined but not fully established, and **[C]** denotes a conjectural goal requiring new mathematical input. When a proved proposition

has explicit unresolved hypotheses, we mark its formal implication [P] and mark the outstanding hypotheses [C] separately, in agreement with the status usage of Part III, Draft 9.

Assertion	Precise basis	Status
Diagonal and nonsplit good-element criterion, including wild p	[FK, §3]; Part I, §3	[E]
Compatibility of the Part I cocycle with σ_{c_v}	Part I, Prop. 2.3(b); direct calculation	[P]
Power-map kernel, index, and Haar modulus on $T_E(F)/F^\times$	Section 4	[P]
Global good-regular class criterion	Section 5; Part III Hasse principle	[P]
Division-side unitary character transform and tame integral case	Part III, Thms. 4.3, 7.6	[E]
Wild rank-two torus Fourier expansions	Worked example in §6.3	[P]
Wild irreducible-character integrality for $\widetilde{D^\times}$	Section 6	[C]
Covered D–covered GL global simple trace comparison	Parts I–II under their hypotheses	[E]
Covered D–ordinary D global comparison	Conditional implication in §8; full matching open	[P]/[C]
Removal of auxiliary support and packet hypotheses	Section 9	[C]

2 The exact cocycle convention and star map

This section fixes the one place where a parameter mismatch could invalidate every subsequent constant. The ambient construction is a restriction of a cocycle over L_v, whereas [FK] uses the standard $\mathrm{GL}_r(F)$ Kazhdan–Patterson convention.

2.1 Restriction and relabeling

Let σ_F be the parameter-zero Kazhdan–Patterson cocycle with restriction to the diagonal torus

$$\sigma_F(\mathrm{diag}(a_i), \mathrm{diag}(b_i)) = \prod_{i<j} (a_i, b_j)_{n,F}. \tag{2}$$

The GL subgroup of the ambient cover in Part I has, *as an extension*, cocycle model

$$\sigma_F(g,h)^r \, (\det g, \det h)_{n,F}^c \quad \text{before relabeling } \mu_n. \tag{3}$$

This assertion is Part I, Proposition 2.3(b). Applying the kernel relabeling $\zeta \mapsto \zeta^{r'}$ yields the model

$$\sigma_{c_v}(g,h) = \sigma_F(g,h)(\det g, \det h)_{n,F}^{c_v}. \tag{4}$$

The two parameters c and c_v must be kept separate: c occurs before relabeling and $c_v = r'c \pmod n$ afterward. The isomorphism in (3) is an isomorphism of extensions, not a claim that arbitrary choices of cocycle sections give literally equal functions on all of $\mathrm{GL}_r(F)^2$.

Proposition 2.1 (**[P]** Diagonal commutator audit)**.** *For $t = \mathrm{diag}(t_1, \dots, t_r)$ and $u = \mathrm{diag}(u_1, \dots, u_r)$ in $\mathrm{GL}_r(F)$,*

$$[\widetilde{t}, \widetilde{u}] = \prod_{i,j} (t_i, u_j)_{n,F}^{A_{ij}}, \qquad A = -I_r + (1 + 2c_v)J_r. \tag{5}$$

Here J_r is the all-ones matrix. Its row sums are $r - 1 + 2rc_v \equiv 0 \pmod n$, so scalar lifts commute with the lifted split torus. The radical on $(F^\times/F^{\times n})^r$ consists precisely of diagonal classes.

Proof. The commutator is the ratio $\sigma_{c_v}(t,u)/\sigma_{c_v}(u,t)$. The Hilbert symbol satisfies $(a,b)_n = (b,a)_n^{-1}$; this is *antisymmetry*, valid for all n. The determinant twist contributes $2c_v$ to each entry of A, and the parameter-zero factor contributes 1 off the diagonal. This gives (5). The Hilbert pairing on $X = F^\times/F^{\times n}$ is perfect. As X is killed by n and $(r,n) = 1$, the decomposition $X^r = \Delta X \oplus \{(x_i) : \sum x_i = 0\}$ holds by averaging with r^{-1} modulo n. On these summands A acts, respectively, by $r - 1 + 2rc_v = 0$ and -1. Therefore $\ker A = \Delta X$. □

Remark 2.2. The assumption $(r,n) = 1$ also follows algebraically from $n \mid r - 1 + 2rc_v$, because any common divisor of r and n must divide 1. We retain it because it is a global hypothesis of the series. The oddness of n is used for the ordinary star-map convention below, not for Hilbert-symbol antisymmetry.

2.2 Star and central characters

For an element h of the ordinary group, choose any lift $\widetilde{h}$ and define $h^* = \widetilde{h}^{\,n}$. It is independent of the lift because μ_n is central. In the present scalar-central cover, for $a \in F^\times$,

$$(ah)^* = a^* h^*, \qquad a^* = \widetilde{a}^{\,n}. \tag{6}$$

If h_1, h_2 commute in an ordinary torus, their lifted commutator is in μ_n, and the class-two power identity

$$(\widetilde{h}_1\widetilde{h}_2)^n = \widetilde{h}_1^{\,n}\widetilde{h}_2^{\,n}[\widetilde{h}_2,\widetilde{h}_1]^{n(n-1)/2} = \widetilde{h}_1^{\,n}\widetilde{h}_2^{\,n} \tag{7}$$

uses n odd. Thus star is multiplicative on the lifted torus. This agrees with the convention in [FK, §4] and [Zou, §3]: Zou writes $x^* = u(x)s(x)^n$, where $u(x) = \pm 1$ is the correction defined in [2, §4] and [3, §3.1]. For odd n the extra sign is absent in the Flicker–Kazhdan convention: the star is the nth power of a lift. The correction $u(x)$ is introduced for the even-n convention; we do not infer that an arbitrary sign automatically lies in μ_n. In the odd-degree setting, therefore, $u(x) = 1$ by definition of the relevant convention (compare [2, §4] and [3, §3.1]). For a scalar $\zeta \in \mu_n(F)$, $\zeta^* = 1$ in the standard diagonal cocycle model: the self-symbol $(\zeta,\zeta)_n = 1$ for odd n, so $s(\zeta I_r)^n = s(I_r) = 1$. This last statement is also invariant under the extension isomorphisms in Part I. For representation characters we first fix a genuine unitary character $\widetilde{\theta}$ of the central preimage $\widetilde{Z}$ and *then* put $\theta(a) = \widetilde{\theta}(a^*)$. The reverse implication is not unique: θ alone does not specify $\widetilde{\theta}$. For $\zeta \in \mu_n(F)$ we have $\zeta^* = 1$, hence $\theta(\zeta) = 1$. Globally the rational splitting gives $a^* = \iota(a^n)$ for $a \in K^\times$; if $\widetilde{\theta}$ is automorphic, it is trivial on the rational splitting, so $\theta(a) = 1$ on $K^\times$. Test functions use the inverse central characters θ^{-1} and $\widetilde{\theta}^{-1}$.

3 Good elements for every maximal torus

A regular maximal torus in $\mathrm{GL}_r(F)$ may be nonsplit even at a place where $D_v \simeq M_r(F)$. This is the situation not treated by the diagonal matrix alone. The following calculation is not new: it is the proof of [FK, §3, Proposition] in the notation of this series, and also underlies Part I, Proposition 3.5.

Theorem 3.1 (**[E]** Flicker–Kazhdan good-element theorem)**.** *Let $E = \prod_{i=1}^k E_i$ be a finite* étale *F-algebra of dimension r, embedded as a regular maximal torus $T_E(F) = E^\times$ in $\mathrm{GL}_r(F)$. Under (1), for any $t \in T_E(F)$ and any lift $\widetilde{t}$,*

$$[\widetilde{t},\widetilde{u}] = (\mathrm{N}_{E/F}\, t, \mathrm{N}_{E/F}\, u)_{n,F}^{1+2c_v} \prod_{i=1}^{k} (t_i, u_i)_{n,E_i}^{-1}. \tag{8}$$

The radical of this commutator pairing is

$$\{t \in T_E(F) : [\widetilde{t}, \widetilde{u}] = 1 \ \forall u \in T_E(F)\} = F^\times T_E(F)^n. \tag{9}$$

For regular t, this is exactly the criterion that $\widetilde{t}$ is good. Both identities are valid when $p \mid n$.

Proof from the cited formula, included for clarity. The commutator formula is [FK, §3], using [KP, Prop. 0.1.5]; its restriction to a single field is exactly Part I, Proposition 3.1 after relabeling [9]. Write $N = \mathrm{N}_{E/F}$. Norm compatibility of local Hilbert symbols gives

$$(Nt, Nu)_{n,F} = \prod_i (Nt, u_i)_{n,E_i}.$$

Thus (8) equals

$$\prod_i ((Nt)^{1+2c_v} t_i^{-1}, u_i)_{n,E_i}.$$

The nondegeneracy of the nth Hilbert symbol over each E_i (including wildly ramified E_i) implies that the commutator is trivial for all u exactly when

$$t_i \equiv (Nt)^{1+2c_v} \pmod{E_i^{\times n}} \quad (1 \le i \le k). \tag{10}$$

This directly says $t \in F^\times E^{\times n}$. Conversely, if $t = ay^n$ for $a \in F^\times$, then $Nt = a^r (Ny)^n$; therefore $(Nt)^{1+2c_v} \equiv a^{r(1+2c_v)} \equiv a \equiv t_i$ modulo $E_i^{\times n}$ by scalar centrality. This proves the criterion. No tameness assumption has been introduced. □

Example 3.2 (The diagonal rank-two check)**.** Take $r = 2$, $n = 3$ and $F = \mathbb{Q}_3(\zeta_3)$. In the notation of Part I, Example 3.7, choose $c = 1$, so $c_v \equiv 2 \pmod 3$. Formula (5) becomes

$$A \equiv \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \pmod 3, \qquad [\widetilde{t}, \widetilde{u}] = (t_1/t_2, u_1/u_2)_{3,F}.$$

Thus $\mathrm{diag}(t_1, t_2)$, when regular, is good exactly when $t_1/t_2 \in F^{\times 3}$. If a quadratic field E/F is embedded as a nonsplit maximal torus, the same theorem says that goodness is equivalent to $t \in F^\times E^{\times 3}$; the two tests are instances of the same formula, not separate conjectures.

Remark 3.3 (Why goodness is not enough for a regular-root formula)**.** The power of a regular element in a *split* torus need not be regular. For $n = 3$ and $F = \mathbb{Q}_3(\zeta_3)$, the element $\mathrm{diag}(1, \zeta_3)$ is regular but not good: $\zeta_3 \notin F^{\times 3}$, since F does not contain ζ_9. Its cube is the identity. A character or orbital-integral formula with $\Delta(t^n)$ in the denominator must restrict to t^n regular (as in Zou's equation (3.2)), or use a separately proved germ extension. On a degree-r *field* torus, by contrast, if t is regular and $t^n \in F$, then Kummer theory gives $[F(t) : F] \mid n$; since $[F(t) : F] = r$ and $(r, n) = 1$, this is impossible. Because r is prime, $F(t^n) = F(t)$ and t^n is regular.

4 Local power-map calculation for étale tori

The key local distinction is between a field centralizer (compact modulo scalars) and a product of fields (not compact modulo scalars). Both occur in the split *group*. The next proposition gives an exact count of the roots and the size of the image at wild and tame places, including the factors that a global trace identity must account for.

Lemma 4.1 ([**P**] Kummer injection for a product algebra)**.** *Let* $E = \prod_{i=1}^{k} E_i$ *be finite* étale *of dimension* r *over* F*, with* $\mu_n \subset F$ *and* $(r, n) = 1$*. Then*

$$F^\times/F^{\times n} \longrightarrow E^\times/E^{\times n} \tag{11}$$

is injective.

Proof. Suppose $a \in F^\times$ is an nth power in every E_i. The Kummer extension $F(a^{1/n})/F$, since $\mu_n \subset F$, is cyclic of degree $d \mid n$ and embeds in every E_i. Therefore $d \mid [E_i : F]$ for every i, so also $d \mid \sum_i [E_i : F] = r$. Coprimality gives $d = 1$ and $a \in F^{\times n}$. □

Proposition 4.2 ([**P**] Kernel, index, and Haar modulus)**.** *Under the hypotheses of Lemma 4.1, put* $A_E = E^\times/F^\times$ *and* $P_n([t]) = [t^n]$*. Then*

$$\ker(P_n) \simeq \mu_n(E)/\mu_n(F) \simeq \mu_n^k/\mu_n, \qquad |\ker(P_n)| = n^{k-1}, \tag{12}$$

$$[A_E : P_n(A_E)] = n^{2(k-1)}|n|_F^{-(r-1)}. \tag{13}$$

In particular, $P_n(A_E)$ *is open and closed. With* any *Haar measure* da *on* A_E *and its restriction to* $P_n(A_E)$*, for compactly supported integrable* q *on that image,*

$$\int_{A_E} q(a^n)\, da = n^{k-1}|n|_F^{-(r-1)} \int_{P_n(A_E)} q(b)\, db. \tag{14}$$

Proof. If $t^n = a \in F^\times$, then a is an nth power in every E_i; by Lemma 4.1, write $a = b^n$ with $b \in F^\times$. Thus each $t_i/b \in \mu_n(E_i) = \mu_n(F)$, giving (12), including its canonical description.

For a local field M containing μ_n the Kummer quotient has

$$|M^\times/M^{\times n}| = n^2|n|_M^{-1}. \tag{15}$$

Indeed the discrete valuation contributes n; the finite torsion unit factor containing μ_n contributes n; and the torsion-free p-adic unit factor contributes $|n|_M^{-1}$ (multiplication by n on its $\mathbb{Z}_p$-lattice). Alternatively, (15) follows from local duality. Since $F^\times/F^{\times n}$ injects into $E^\times/E^{\times n}$, the group A_E/A_E^n is its cokernel:

$$|A_E/A_E^n| = \frac{\prod_i n^2|n|_{E_i}^{-1}}{n^2|n|_F^{-1}} = n^{2(k-1)}|n|_F^{-(r-1)},$$

where $|n|_{E_i} = |n|_F^{[E_i:F]}$ for normalized absolute values on local fields. This proves (13).

For the Haar statement, write $K_0 = \ker(P_n)$, of order n^{k-1}. Choose a sufficiently small compact open subgroup $U \subset A_E$ consisting of principal-unit classes and having $U \cap K_0 = \{1\}$. The p-adic logarithm identifies U with a free $\mathbb{Z}_p$-module of rank $(r-1)[F : \mathbb{Q}_p]$: the factors $E_i^\times$ have combined principal-unit rank $r[F : \mathbb{Q}_p]$, and quotienting by the scalar principal units subtracts $[F : \mathbb{Q}_p]$. Multiplication by n maps U injectively into itself, with

$$[U : U^n] = p^{v_p(n)(r-1)[F:\mathbb{Q}_p]} = |n|_F^{-(r-1)}.$$

Since $P_n^{-1}(U^n) = K_0U$ and $K_0 \cap U = \{1\}$, its pushforward of Haar measure gives

$$\frac{\mathrm{vol}(P_n^{-1}(U^n))}{\mathrm{vol}(U^n)} = \frac{|K_0|\,\mathrm{vol}(U)}{\mathrm{vol}(U)/[U : U^n]} = n^{k-1}|n|_F^{-(r-1)}.$$

The pushforward is translation-invariant on the open image $P_n(A_E)$, so this ratio is its constant density on that image, proving (14). Notice that this argument requires neither a global quotient-index formula nor a choice of scale for Haar measure. □

Remark 4.3 (A measure-theoretic warning)**.** For a general open endomorphism $\varphi : A \to A$ of a locally compact abelian group with finite kernel, its pushforward Haar density need not equal $[A : \varphi A]/|\ker \varphi|$. For example, on $A = \mathbb{Z}$ with counting measure, $a \mapsto na$ has density 1 on $n\mathbb{Z}$, not n. The correct local computation here is $|\ker P_n|[U : U^n]$, where U is the compact open subgroup in the proof. For this specific A_E it happens to simplify to $[A_E : P_n A_E]/|\ker P_n|$; that numerical equality is not a general change-of-variables theorem.

Example 4.4 (Three local torus types)**.** If E/F is a field of degree r, then $k = 1$ and P_n is injective, with index and Haar modulus $|n|_F^{-(r-1)}$. If $E = F^r$ then $k = r$ and the kernel has order n^{r-1}. For $r = 2, n = 3$, $F = \mathbb{Q}_3(\zeta_3)$ and $|3|_F = 1/9$,

type	$\lvert\ker P_3\rvert$	$[A_E : A_E^3]$	factor in (14)
E/F quadratic	1	9	9
$E = F \times F$	3	81	27

These integers quantify why one cannot use the division-torus substitution formula unchanged on the full split group.

5 A global classification of good regular classes

The local theorem now applies to the actual torus at *every* place. Combining it with the Hasse principle of Part III removes the local-to-global ambiguity for power classes of a degree-r field, though not the independent measure and spectral questions.

Proposition 5.1 ([**E**] Almost-everywhere powers modulo scalars)**.** *Let E/K be a field extension of degree r with $(r, n) = 1$ and $\mu_n \subset K$. For every finite set Σ of places,*

$$E^\times/(K^\times E^{\times n}) \longrightarrow \prod_{v \notin \Sigma} \frac{(E \otimes_K K_v)^\times}{K_v^\times (E \otimes_K K_v)^{\times n}} \tag{16}$$

is injective. This is Part III, Proposition 11.3 [11].

Recalled proof. Let t be locally scalar times an nth power outside Σ. Choose $u, k \in \mathbb{Z}$ with $ru+nk = 1$ and put $a = N_{E/K}(t)^u \in K^\times$. If $t = a_v y_v^n$ in the local étale algebra, taking norms gives the identity

$$t/a = \left(a_v^k y_v / N_{E_v/K_v}(y_v)^u\right)^n.$$

Thus t/a is an nth power in every completion of E over all $v \notin \Sigma$. If its class in $E^\times/E^{\times n}$ were nontrivial, the nontrivial Kummer extension $E((t/a)^{1/n})/E$ would, by Chebotarev, admit infinitely many unramified places at which t/a is not an nth power. This contradicts the hypothesis. Hence $t \in K^\times E^{\times n}$. □

Theorem 5.2 ([**P**] Global good-regular criterion)**.** *Assume the covers and rational splittings of Part I. Let G denote $D^\times$ or GL_r, and let $\gamma \in G(K)$ be regular elliptic, so $E = K[\gamma]$ is a field of degree r. The following are equivalent:*

(a) the rational lift $\iota(\gamma)$ is good at every place;

(b) it is good at every place outside some finite set Σ;

(c) $\gamma \in K^\times E^{\times n}$.

The map $[s] \mapsto [s^n]$ is a bijection from the noncentral classes in $E^\times/K^\times$ onto the globally good regular classes belonging to E. Furthermore

$$W_{s^n} = W_s, \qquad W_s = \{\sigma \in \mathrm{Aut}(E/K) : \sigma(s) \in K^\times s\}. \tag{17}$$

Proof. At each finite v, the ordinary centralizer is $T_v = (E \otimes_K K_v)^\times$. If D_v splits, apply Theorem 3.1, transported through Part I, Proposition 2.6 (the split-place isomorphism of the specified covers). If D_v is division, use Part I, Proposition 3.5(a), whose commutator is the same by Part I, Proposition 3.1. At complex places the nth-power map on T_v is surjective. Thus at every place the local goodness condition is precisely $\gamma \in K_v^\times T_v^n$. Implications (c)⇒(a)⇒(b) follow, while (b)⇒(c) is Proposition 5.1.

The power map $E^\times/K^\times \to E^\times/K^\times$ is injective: if $s^n \in K^\times$, Kummer theory shows $[K(s) : K] \mid n$, while $[K(s) : K] \mid [E : K] = r$. Hence $s \in K^\times$. Because r is prime, every noncentral $s \in E$ generates E, and the same is true of s^n. This proves the bijection. Finally $\sigma(s^n)/s^n \in K^\times$ if and only if $(\sigma(s)/s)^n \in K^\times$, which is equivalent by injectivity to $\sigma(s)/s \in K^\times$. Equation (17) follows. □

Remark 5.3 (Precise consequence for the geometric side)**.** Part I, Proposition 6.1, uses the rational regular classes which are good everywhere, with coefficient

$$c(\gamma) = \frac{\mathrm{vol}(\mathbb{A}_K^\times E^\times \backslash \mathbb{A}_E^\times; dt)}{|W_\gamma|}. \tag{18}$$

Theorem 5.2 parametrizes its noncentral summation set by global $[s^n]$ and preserves W_γ. It does *not* replace the local root sums in Zou's character formula: the global root class may be unique even when a localized split torus has n^{k_v-1} roots modulo scalars.

6 The remaining wild local character problem

We distinguish character transfer from class-function transfer. The latter is a theorem of Part III; the former requires an *integral* statement not implied by unitarity.

6.1 Division-side normalization and discriminants

Let F be nonarchimedean, D/F a division algebra of prime degree r, $H = D^\times$, and $\widetilde{H}$ its specified cover. Set $Z = F^\times$ and $\widetilde{Z} = p^{-1}(Z)$, equipped with compatible Haar measures on $H/Z \simeq \widetilde{H}/\widetilde{Z}$. The Part III transform is

$$(L_D\Phi)(z\,t^*) = b_D\,\widetilde{\theta}(z)\frac{\Delta(t)}{\Delta(t^n)}\Phi(t), \qquad b_D = |n|_F^{-(r-1)/2}, \tag{19}$$

for regular t, $z \in \widetilde{Z}$; it vanishes at nongood regular lifts. Here

$$\Delta(t) = \left|\det(1 - \mathrm{Ad}(t) \mid \mathfrak{h}/\mathfrak{t})\right|_F^{1/2} = \prod_{i<j}\left|\frac{(\lambda_i - \lambda_j)^2}{\lambda_i\lambda_j}\right|_F^{1/2}. \tag{20}$$

This equals Part I's $|D(t)|^{1/2}|\,\mathrm{Nrd}(t)|_F^{-(r-1)/2}$. Since the power map on a field torus modulo scalars has index $|n|_F^{-(r-1)}$, Weyl integration gives b_D^2 equal to this index. The factor b_D is independent of

common rescalings of the two identified quotient Haar measures. The ratio of discriminants extends at central elements to $|n|_F^{-r(r-1)/2}$ (Part III, Lemma 3.3), hence

$$(L_D\Theta_\rho)(1) = c_D \dim \rho, \qquad c_D = b_D|n|_F^{-r(r-1)/2} = |n|_F^{-(r^2-1)/2}. \tag{21}$$

The identity factor is pointwise and does not depend on Haar measures. Part III, Theorem 9.1, separately gives the adjoint orbital-integral/test-function transfer. Test functions transform by $\widetilde{\theta}^{-1}$, not by $\widetilde{\theta}$.

6.2 The integral matrix

Choose $\widetilde{\theta}$ first and put $\theta(a) = \widetilde{\theta}(a^*)$. Part III, Theorem 4.3 and Corollary 5.1, show that L_D is a unitary *isomorphism* of invariant L^2 spaces and that

$$L_D\Theta_\rho = \sum_\tau a_{\rho\tau}\Theta_\tau, \qquad a_{\rho\tau} = \langle L_D\Theta_\rho, \Theta_\tau\rangle, \tag{22}$$

with each row and column finite. The characters on either side are orthonormal bases in their fixed-central-character spaces.

Proposition 6.1 ([**E**] Integral-lattice criterion, Part III)**.** *The transform L_D permutes the irreducible characters with positive signs if and only if every $a_{\rho\tau}$ in (22) is an integer.*

Proof. An integral unit vector in ℓ^2 is a signed basis vector; unitarity gives injectivity and surjectivity because the inverse is the adjoint of a unitary map. The value at identity in (21) is positive and fixes the sign. This is Part III, Corollary 5.2, with its converse. □

Remark 6.2 (The actual obstruction)**.** For wild v, Part III does not prove $a_{\rho\tau} \in \mathbb{Z}$. The tame Adams argument works with a finite quotient Q and a central subgroup C, under the hypothesis $\gcd(n, |Q/C|) = 1$ (Part III, Lemma 7.5). At a wild place that hypothesis need not hold. For the simplest illustration take $Q = C_p$, $C = 1$, $n = p$: then the Adams operation sends every linear character to the trivial character. This does not prove a nonintegral coefficient for L_D; it only shows why that particular proof method fails.

6.3 A fully worked wild example: all quadratic field tori

The following calculation is more informative than the dimension obstruction alone. It verifies a *necessary representation-ring integrality test on every maximal torus*, including the two ramified quadratic field types; the full-group integrality statement remains open.

Example 6.3 (The field, algebra, and cover)**.** Take

$$F = \mathbb{Q}_3(\zeta_3), \qquad \varpi = \zeta_3 - 1, \qquad 3 = -\zeta_3^{-1}\varpi^2, \qquad q_F = 3.$$

Thus $v_F(3) = 2$ and $|3|_F = 3^{-2} = 1/9$. The compatible covering parameters in [JC] and Part I are respectively $c = 1$ and $c_v = 2 \pmod 3$. Let D/F be the unique quaternion division algebra. With $E_0 = F(i)$, $i^2 = -1$, its unramified quadratic maximal subfield, we may write

$$D = E_0 \oplus E_0 j, \qquad j^2 = \varpi, \qquad jx = \sigma(x)j \quad (x \in E_0),$$

where σ is the nontrivial automorphism of E_0/F. Part I, Proposition 3.5, gives, for every quadratic maximal subfield E/F, the covered-torus commutator

$$[x, y]_{\widetilde{E^\times}} = (N_{E/F}x, N_{E/F}y)_{F,3}^2 (x, y)_{E,3}^{-1}. \tag{23}$$

The center C_E of $\widetilde{E^\times}$ is the preimage of $F^\times E^{\times 3}$, and

$$[E^\times : F^\times E^{\times 3}] = 9, \qquad b_D = 3, \qquad c_D = 27.$$

By the Heisenberg-type Stone–von Neumann theorem ([JC], Proposition 2.4; Part I, Proposition 3.5), every genuine irreducible representation of $\widetilde{E^\times}$ has degree 3. This is *not* a degree calculation for representations of $\widetilde{D^\times}$.

Lemma 6.4 (A general toral Fourier test)**.** *Let E/F be any maximal field in D, with r prime and $(r, n) = 1$, and put $A_E = E^\times/F^\times$. The function*

$$R_E(t) = \frac{\Delta(t)}{\Delta(t^n)}$$

extends continuously at scalars, is a locally constant function on the compact abelian group A_E, and has a finite Fourier expansion

$$R_E(t) = \sum_{\chi \in \widehat{A_E}} m_\chi \chi(t), \qquad m_\chi = \int_{A_E} R_E(t)\overline{\chi(t)}\, dt. \tag{24}$$

Here the Haar measure of A_E has mass 1. Choose a compatible genuine central character $\widetilde{\theta}$ of $\widetilde{Z}$ and set $\theta(a) = \widetilde{\theta}(a^*)$. For any smooth character $\alpha : E^\times \to \mathbb{C}^\times$ with $\alpha|_{F^\times} = \theta$, and for each $\chi \in \widehat{A_E}$, let $\eta_{\alpha\chi}$ be the unique genuine irreducible representation of $\widetilde{E^\times}$ whose central character satisfies

$$\zeta_{\alpha\chi}(z\, t^*) = \widetilde{\theta}(z)\alpha(t)\chi(t) \qquad (z \in \widetilde{Z},\ t \in E^\times).$$

Define a torus-level class function $\mathcal{L}_E\alpha$ to equal $b_D\widetilde{\theta}(z)R_E(t)\alpha(t)$ on $zt^* \in C_E$ and to vanish outside C_E. Then, as a genuine class function on the covered torus*,*

$$\mathcal{L}_E\alpha = \sum_\chi m_\chi\, \Theta_{\eta_{\alpha\chi}}. \tag{25}$$

For an ordinary character Θ_ρ of $D^\times$, its actual L_D transform restricts to the sum of these $\mathcal{L}_E\alpha$ with the multiplicities in $\Theta_\rho|_{E^\times}$. Consequently $m_\chi \in \mathbb{Z}_{\geq 0}$ for every χ is sufficient for the restriction of every transferred ordinary irreducible character to this covered torus to be a genuine representation character. It is not sufficient for the full $\widetilde{D^\times}$ character to be a representation character.

Proof. The extension of R_E at scalars follows from Part III, Lemma 3.3. Since E/F is a degree-r field extension with $(r, n) = 1$, the nth power of a noncentral t stays noncentral. The compactness of A_E and local constancy imply that R_E factors through a finite quotient, giving (24).

By Part I the center of the covered torus is $C_E = \widetilde{Z} \cdot \{t^* : t \in E^\times\}$. The displayed formula for $\zeta_{\alpha\chi}$ is well-defined: the power map on A_E is injective, and on scalar overlaps the compatibility $\alpha(a) = \widetilde{\theta}(a^*)$ is precisely what is required. Such genuine central characters extend over C_E, and Stone–von Neumann gives a unique $\eta_{\alpha\chi}$. Its character vanishes off C_E, while on C_E it equals $b_D\zeta_{\alpha\chi}$ because its degree is b_D. The defining formula (19) for L_D has the identical vanishing and equals $b_D\widetilde{\theta}(z)R_E(t)\alpha(t)$ on zt^*. Comparing with (24) proves (25). Finally, restricting a finite-dimensional ordinary representation to $E^\times$ decomposes it into one-dimensional characters, all restricting to θ on $F^\times$; summing the displayed formula proves the last claim. □

6.3.1 The unramified quadratic torus

Put $E_0 = F(i)$ as above. Since $X^2 + 1$ is irreducible over $\mathbb{F}_3$, E_0/F is unramified and its residue field is $\mathbb{F}_9$. For $t = a + bi$ with $a, b \in F$ the reduced norm and trace give the elementary identity

$$R_{E_0}(a + bi) = \frac{|a^2 + b^2|_F}{|3a^2 - b^2|_F}. \tag{26}$$

Multiplying t by an element of $F^\times$, suppose $a, b \in \mathcal{O}_F$ and at least one is a unit. Then $a^2 + b^2$ is always a unit. If b is a unit, so is $3a^2 - b^2$. If $b = \varpi u$, then

$$3a^2 - b^2 = 3(a^2 + \zeta_3 u^2),$$

and a is a unit; the reduction of $a^2 + \zeta_3 u^2$ modulo ϖ is either 1 or 2, never 0. Hence

$$R_{E_0}(a + bi) = \begin{cases} 9, & b \in \varpi\mathcal{O}_F, \\ 1, & b \in \mathcal{O}_F^\times. \end{cases} \tag{27}$$

The finite residue quotient is

$$B_0 = k_{E_0}^\times / k_F^\times \simeq C_4.$$

The condition $b \in \varpi\mathcal{O}_F$ says exactly that the image of t in B_0 is the identity. Denote its four characters by $\chi_0 = 1, \chi_1, \chi_2, \chi_3$. The regular character of C_4 equals 4 at the identity and 0 elsewhere. Thus

$$\boxed{R_{E_0} = 1 + 2\,\mathrm{reg}_{C_4} = 3\chi_0 + 2\chi_1 + 2\chi_2 + 2\chi_3.} \tag{28}$$

This is a positive integral Fourier decomposition of degree $3 + 2 + 2 + 2 = 9$; it is stronger than merely checking that R_{E_0} takes integer values.

6.3.2 Both ramified quadratic tori

Every ramified quadratic extension of F can be written $E_d = F(j_d)$, $j_d^2 = d = u\varpi$ with $u \in \mathcal{O}_F^\times$. There are two isomorphism classes, distinguished by the square class of u. For $t = a + bj_d$,

$$R_{E_d}(a + bj_d) = \frac{|a^2 - db^2|_F}{|3a^2 + db^2|_F}. \tag{29}$$

Again normalize a, b to be integral and not both divisible by ϖ. The terms $3a^2$ and db^2 have different valuations whenever both a and b are units. A direct valuation calculation therefore gives

$$R_{E_d}(a + bj_d) = \begin{cases} 9, & a \in \mathcal{O}_F^\times, \quad b \in \varpi\mathcal{O}_F, \\ 3, & a, b \in \mathcal{O}_F^\times, \\ 1, & a \in \varpi\mathcal{O}_F, \quad b \in \mathcal{O}_F^\times. \end{cases} \tag{30}$$

For example, in the original cyclic algebra $d = \varpi$,

t	j	$1 + j$	$1 + \varpi j$
$R(t)$	1	3	9
$(L_D 1)(t^*)$	3	9	27

for the central-character normalization of Example 6.3. The three values illustrate why the unramified torus alone is not an adequate test.

There is a natural finite quotient

$$B_d = E_d^\times/(F^\times U_1) \simeq C_2 \times C_3,$$

where $U_1 \subset \mathcal{O}_{E_d}^\times$ consists of $a + bj_d$ with $a \in \mathcal{O}_F^\times$ and $b \in \varpi\mathcal{O}_F$. Modulo $F^\times$ its members are represented by $1 + cj_d$ with $c \in \varpi\mathcal{O}_F$. The C_2 coordinate records the parity of the E_d-valuation, and on even elements the C_3 coordinate is $\bar{c} \in \mathbb{F}_3$ for $1 + cj_d$. The multiplication law

$$(1 + cj_d)(1 + c'j_d) = (1 + dcc')\left(1 + \frac{c + c'}{1 + dcc'}j_d\right)$$

shows that $\bar{c}$ is additive, and $j_d^2 = d \in F^\times$ provides the order-two factor.

Write ϵ for the nontrivial character of C_2 and ξ for a character of order 3 of C_3. The identity (30) says that R_{E_d} has values 9 at the identity of B_d, 3 at its two nonidentity elements in C_3, and 1 on its three odd elements. Consequently

$$\boxed{\begin{aligned} R_{E_d} &= 1 + \mathrm{reg}_{C_2} + \mathrm{reg}_{C_2 \times C_3} \\ &= 3 \cdot 1 + 2\epsilon + \xi + \xi^2 + \epsilon\xi + \epsilon\xi^2. \end{aligned}} \tag{31}$$

Here the regular representations are inflated from the indicated quotients. All coefficients are nonnegative integers and sum to $3 + 2 + 1 + 1 + 1 + 1 = 9$. This applies to *both* ramified quadratic isomorphism classes, independently of u.

Proposition 6.5 ([**P**] All quadratic torus restrictions pass an integral-character test)**.** *In the wild example $F = \mathbb{Q}_3(\zeta_3)$, $r = 2$, $n = 3$, let D/F be the quaternion division algebra. The function R_E is the character of a nine-dimensional permutation representation of a finite quotient of $E^\times/F^\times$, for every quadratic maximal subfield $E \subset D$. More precisely, for unramified E it is $1 \oplus 2\,\mathbb{C}[C_4]$, and for either ramified E it is $1 \oplus \mathbb{C}[C_2] \oplus \mathbb{C}[C_2 \times C_3]$. For a suitable genuine central character inducing the trivial ordinary central character, the restriction of $L_D\mathbf{1}$ to $\widetilde{E^\times}$ is therefore an* actual genuine representation character *of total dimension* 27*, namely*

E/F	*multiplicities of three-dimensional genuine torus irreducibles*
unramified	$(3, 2, 2, 2)$
ramified, either type	$(3, 2, 1, 1, 1, 1)$.

These multiplicities are invariant under the nontrivial F-automorphism of each E. No irreducibility or even integrality for the full $\widetilde{D^\times}$ is asserted.

Proof. A p-adic field of odd residue characteristic has four square classes; hence F has exactly three quadratic field extensions, one unramified and two ramified. Every quadratic field extension embeds in the quaternion division algebra. Equations (28) and (31) prove the first assertion. Lemma 6.4, together with the degree-three torus Stone–von Neumann theorem, converts those Fourier coefficients into the displayed multiplicities; their weighted dimensions are $3 \cdot 9 = 27$. For the trivial ordinary character one can choose $\widetilde{\theta}$ with $\widetilde{\theta}(a^*) = 1$: the odd-$n$ scalar star kills $\mu_3 \subset F^\times$ (Section 2.2), and a faithful character of μ_3 extends across the abelian scalar central group modulo the star image. The nontrivial Galois automorphism acts by inversion on the respective finite quotients' character parameters (with the C_2 factor fixed), so the displayed multiplicities are invariant. □

6.4 From toral positivity to the general problem

The example suggests two distinct conjectural steps, and shows why the first alone is insufficient.

Step T: positivity on each field torus. For a general prime degree r, odd n and wild F, test whether R_E has a nonnegative integral Fourier expansion on $E^\times/F^\times$ for *every* degree-r field E/F. A particularly tractable sufficient condition is to write R_E as a nonnegative integral combination of *permutation characters* of finite abelian quotients:

$$R_E(t) = \sum_{U \le A} c_U [A : U]\, \mathbf{1}_U(t), \qquad c_U \in \mathbb{Z}_{\ge 0}, \tag{32}$$

where A is a finite quotient of $E^\times/F^\times$ and U ranges over its subgroups. Indeed $[A : U]\, \mathbf{1}_U = \mathrm{Ind}_U^A \mathbf{1}$ is a permutation character for abelian A. The two formulas above are explicit examples of (32). To seek such expressions generally, use the tower of unit filtrations of E/F and the identity

$$R_E(t) = \left| \det\left(\sum_{j=0}^{n-1} \mathrm{Ad}(t)^j \;\middle|\; \mathfrak{d}/\mathfrak{e} \right) \right|_F^{-1/2}, \tag{33}$$

where $\mathfrak{d} = D$ and $\mathfrak{e} = E$ as F-Lie algebras; this follows from $1 - X^n = (1 - X)(1 + X + \cdots + X^{n-1})$ on the noncentral adjoint space. The finite valuation strata in (33) are candidates for indicators of congruence subgroups. Neither their integrality nor their positivity is established for general r, n, F.

Step G: nonabelian gluing. Even if Step T holds on every maximal field torus, the local restrictions must arise *simultaneously* from one character of $\widetilde{D^\times}$. The ramified example exposes extra congruence data invisible in the unramified residual quotient: $R(j) = 1$, $R(1 + j) = 3$, and $R(1 + \varpi j) = 9$. Torus positivity does not impose the compatibility of the corresponding Heisenberg models under conjugation by $D^\times$ or supply the integral coefficients of the full character matrix (22).

A concrete test is to choose the maximal order $\mathcal{O}_D$, its radical $\mathfrak{P}_D$, and $U_D^m = 1 + \mathfrak{P}_D^m$; select an m sufficiently large that the relevant smooth class functions and representations are controlled by a finite level. One must retain the genuine central character in a *twisted* finite group algebra, rather than quotienting out the genuine center. At each level: (1) calculate the induced cocycle and genuine irreducibles; (2) use Clifford theory for the normal congruence subgroups; (3) compare their restrictions to all quadratic covered tori; (4) compute the actual row coefficients $\langle L_D \mathbf{1}, \Theta_\tau \rangle$ exactly. A single noninteger coefficient disproves full integrality; integer coefficients at one level are not a uniform proof for all smooth irreducibles. In the general case one must then establish level-independent compatibility for arbitrary ordinary ρ, not only $\rho = \mathbf{1}$.

7 Connection with Zou's corrected character relation

For the Kazhdan–Patterson cover of the *split group*, Zou [3, §3] revisits the Flicker–Kazhdan metaplectic correspondence and records a corrected scalar in the character relation. This is an essential input, but its objects are $\widetilde{\mathrm{GL}_r(F)}$ and ordinary $\mathrm{GL}_r(F)$, not directly $\widetilde{H}$ and H at a division place.

Proposition 7.1 (**[E]** Zou's root-sum constant in our case)**.** *Write $d_F(m) = m|m|_F^{-1/2}$ and $d_r = \gcd(n, 2rc_v + r - 1)$. Under (1), $d_r = n$. In Zou's equation (3.2), the coefficient preceding the root sum in rank r is*

$$\frac{1}{d_F(d_r)|n|_F^{r/2}} = \frac{1}{n|n|_F^{(r-1)/2}} = \frac{b_D}{n}. \tag{34}$$

For regular elliptic x with degree-r field centralizer, x^n is automatically regular by Remark 3.3. The equation $t^n = x^n$ then has exactly n solutions $t = \zeta x$, $\zeta \in \mu_n$. If the linear character π has central

character ω_π trivial on μ_n, these summands agree and the effective coefficient is b_D. This restriction on ω_π is essential: without it, the scalar sum is $\sum_{\zeta\in\mu_n}\omega_\pi(\zeta)=0$, so equation (3.2) instead forces the covered character to vanish at x^*. For a genuine lift pair the compatibility $\omega_\pi(a)=\omega_{\widetilde{\pi}}(a^*)$ and $\zeta^*=1$ imply $\omega_\pi|_{\mu_n}=1$ automatically.

Proof. The equation $d_r = n$ is immediate from scalar centrality; (34) follows by substitution. Regularity of x^n follows from the field case of Remark 3.3. Any solution t centralizes x^n, so $t \in F[x]^\times$. In the field $F[x]$, $(t/x)^n = 1$, and $\mu_n(F[x]) = \mu_n(F)$. Thus $t = \zeta x$. The Weyl discriminant is invariant under central scalar multiplication, the ordinary character takes the same value because it is trivial on μ_n, and $(\zeta x)^* = x^*$ by the explicit odd-n scalar star computation in Section 2.2. These are precisely the equal terms in Zou's (3.2). □

Remark 7.2 (What is and is not composed)**.** At the ramified places one may compare the *desired* $\overline{G_{D_v}} \leftrightarrow \overline{G}_v$ character relation of Parts I–II with (a) the classical linear Jacquet–Langlands elliptic character identity and (b) Zou's formula. The common coefficient b_D and the familiar sign $(-1)^{r-1}$ are consistent with this comparison. Nevertheless *the existence of an irreducible genuine image on the covered division side is the missing conclusion*; combining two known formulas cannot be used to assume that missing representation exists. For non-elliptic split centralizers, retain Zou's entire root sum, including the central-character factors.

8 Global geometric comparison: proved input and new obligations

There are two comparisons to keep distinct:

$$\begin{aligned} \overline{G_D}(\mathbb{A}_K) &\longleftrightarrow \overline{G}(\mathbb{A}_K) \quad \text{(Parts I–II, under their hypotheses)},\\ D^\times(\mathbb{A}_K) &\dashrightarrow \overline{G_D}(\mathbb{A}_K) \quad \text{(proposed additional comparison)}. \end{aligned} \tag{35}$$

The first has a proved simple trace-formula comparison with regular-support or later matching hypotheses (Part I, Theorem 6.6; Part II, Theorem 6.2). The second is not a formal consequence of the first or of the unitarity of L_D.

8.1 Geometric coefficients and the product formula

For a regular elliptic rational γ, with $E = K[\gamma]$, Part I identifies the torus coefficient $c(\gamma)$, recalled in (18). The same torus measure dt must be used whenever this term is compared with the corresponding linear inner-form orbital integral. Part I uses the normalized orbital integral $J_v(\gamma, f) = \Delta_v(\gamma)O_v(\gamma, f)$. For a rational regular element the product formula implies

$$\prod_v \Delta_v(\gamma) = 1. \tag{36}$$

This is because each discriminant is the normalized absolute value of a nonzero element of K (or its square root), including the determinant normalization in (20). The local *ratio* of discriminants $\Delta_v(t)/\Delta_v(t^n)$ must be kept until local transfer and the global product are both defined.

At $v \in S$ Part III, Theorem 9.1, gives the exact local normalized orbital-integral identity

$$J_{\widetilde{H}_v}(z\,t^*, \mathcal{O}_{D,v}f_v) = b_{D,v}\,\widetilde{\theta}_v(z)^{-1}J_{H_v}(t, f_v), \tag{37}$$

using the same quotient measures on the two copies of H_v/Z_v. At $v \notin S$, $H_v \cong \mathrm{GL}_r(K_v)$ and the appropriate root-sum comparison is not (37); it must use Flicker–Kazhdan–Zou, and it can involve the kernel $\mu_n^{k_v}/\mu_n$ of Proposition 4.2. In particular the local factors $b_{D,v}$ for $v \in S$ *cannot* simply be cancelled using $\prod_v |n|_v = 1$ without the split-side contributions.

8.2 Haar ledger

For $v \in S$, Part II, Definition 2.1, fixes

$$\mathrm{vol}(D_v^\times / F_v^\times; dg_{D,v}) = d(\mathrm{St}; dg_{\mathrm{GL},v})^{-1}, \tag{38}$$

where $d(\mathrm{St}; dg_{\mathrm{GL},v})$ is the formal degree relative to the chosen $\mathrm{GL}_r(F_v)/F_v^\times$ measure. The probability measure used in Part III is $d(\mathrm{St}; dg_{\mathrm{GL},v})\, dg_{D,v}$. When *both* sides of (37) are scaled by the same measure, $b_{D,v}$ is unchanged. Different measures on the GL side must still be reconciled with (38).

Item	Required statement or calculation	Status
Ambient cocycle and common center	Part I, Lemma 5.1 and Prop. 5.5; restriction (3)	[E]
Regular rational good locus	Theorem 5.2	[P]
Global rational/stable classes	Hilbert 90 and Kottwitz's framework; local embeddability into D	[E]
Global torus coefficient	Same dt in (18), including $\lvert W_\gamma \rvert$	[E]
Local $v \in S$ orbital transfer	Part III, Theorem 9.1, equation (37)	[E]
Split-place $v \notin S$ invariance	Lemma 8.2 (necessary); rank-two spherical calculation below	[P]
Split-place pointwise transfer and λ_v	Proposition 8.5; unproved existence	[C]
Unramified unit elements	[2, §§11–12] under $(n, N_r) = 1$; transport to Part I cover pending	[E]/[O]
Nonregular powers and singular germs	Remark 3.3; germ compatibility or support needed	[C]
Central identity class	Suppressed in Prop. 8.5; separate germ/measure test below	[E]/[C]
Spectral expansion	Part I–II packet formula with restrictions; new factor matching	[E]/[C]

Here N_r is the least common multiple of the *composite* positive integers at most r (the convention in [FK]), with $N_r = 1$ when there are none. The classical unramified unit matching in [2, §12, Corollary] is established under $(n, N_r) = 1$ and $p \nmid n$. Thus $N_2 = N_3 = 1$, and for prime $r = 5$ we have $N_5 = 4$, automatically prime to the odd integer n. Other ranks must retain the published hypothesis.

8.3 Split-place invariance and a geometric comparison criterion

At a place $v \notin S$ the algebra D_v is split, but a rational elliptic field $K[s]$ may decompose locally as a product of fields. By Proposition 4.2, its local nth-root fibre modulo scalars can contain $n^{k_v - 1}$ points. The global class $[s]$ supplied by Theorem 5.2 is nevertheless unique. Therefore a local sum over *all* roots cannot be substituted for the orbital integral of this one rational representative without an additional compatibility statement.

Definition 8.1 (FK-compatible ordinary functions)**.** Fix the local central characters $\theta_v, \widetilde{\theta}_v$ with $\theta_v(a) = \widetilde{\theta}_v(a^*)$. A normalized ordinary test function f_v is called *FK-compatible on the regular-power*

locus if, whenever $x, x' \in H_v$ have x^n, x'^n regular and their star images satisfy $(x')^* = zx^*$ for a central lift $z \in \widetilde{Z}_v$ (after conjugating representatives), one has

$$J_{H_v}(x', f_v) = \widetilde{\theta}_v(z)^{-1} J_{H_v}(x, f_v). \tag{39}$$

The case $x' = \xi x$ with *scalar* $\xi \in \mu_n(F)$ is automatic, since $\theta_v(\xi) = 1$. The substantive condition occurs on a product torus $E_v = \prod_{i=1}^{k_v} E_i$ with $k_v \geq 2$: if $\zeta = (\zeta_i) \in \mu_n(E_v) \simeq \mu_n(F)^{k_v}$ is nonscalar and x^n is regular, then

$$J_{H_v}(\zeta x, f_v) = J_{H_v}(x, f_v). \tag{40}$$

The odd-n star is multiplicative on each lifted torus, and $\zeta^* = 1$, so $(\zeta x)^* = x^*$. The kernel on the quotient torus is $\mu_n(F)^{k_v}/\mu_n(F)$, of size n^{k_v-1} (Proposition 4.2). Equation (39) is the invariant-function condition of [2, §6], expressed with the present inverse central transformation convention.

Lemma 8.2 ([**P**] Necessary split-place invariance)**.** *Suppose a genuine $\widetilde{f}_v$ and a fixed nonzero constant λ_v satisfy*

$$J_{\widetilde{H}_v}(x^*, \widetilde{f}_v) = \lambda_v J_{H_v}(x, f_v) \tag{41}$$

for all x in a conjugation- and scalar-stable region where x^n is regular, with the identity at central translates understood with the prescribed inverse-central-character law. Then f_v obeys (39) on that region.

Proof. If $(x')^* = zx^*$, conjugation invariance and the inverse central transformation law of the genuine test function imply $J_{\widetilde{H}_v}((x')^*, \widetilde{f}_v) = \widetilde{\theta}_v(z)^{-1} J_{\widetilde{H}_v}(x^*, \widetilde{f}_v)$. Apply (41) twice and divide by the fixed nonzero λ_v. In particular, for $x' = \zeta x$ with nonscalar $\zeta \in \mu_n(E_v)$, one has $(x')^* = x^*$ and obtains (40). The scalar-root case is already automatic from the central-character convention. Thus FK-compatibility is a *necessary* condition, not an asserted construction of $\widetilde{f}_v$. □

Proposition 8.3 ([**P**] Rank-two tame spherical-unit check)**.** *Let F be nonarchimedean with $\mu_n \subset F$, $p \nmid n$, $r = 2$, and $K_0 = \mathrm{GL}_2(\mathcal{O}_F)$. Take a spherical ordinary unit f^0, supported on $F^\times K_0$, with an unramified central character, and the standard quotient measures. Then f^0 obeys (40) for all regular split-torus x with x^n regular. For a quadratic field torus $E^\times$, the root-fibre condition modulo scalars is vacuous. Moreover, when f^0 and the genuine spherical unit $\widetilde{f}^0$ are transported from the standard Flicker–Kazhdan cocycle and measure conventions, [2, §12, Corollary] establishes their regular-power matching with $\lambda_v = 1$ in that normalization. Whether the same equality holds* literally *for the selected ambient cover and global measures of Parts I–IV requires their explicit splitting and Haar identifications.*

Proof. A split regular element is conjugate to $\mathrm{diag}(a, b)$; multiplying by a scalar, first consider $a, b \in \mathcal{O}_F^\times$. The Iwasawa and unipotent-orbit calculation, with $\mathrm{vol}(\mathcal{O}_F) = 1$, yields

$$O_{\mathrm{GL}_2}(\mathrm{diag}(a, b), 1_{K_0}) = |a - b|_F^{-1}, \qquad \Delta(\mathrm{diag}(a, b)) = |a - b|_F,$$

so $J = 1$. Multiplication of either eigenvalue by a root of unity preserves its valuation, hence the normalized orbital integral. If the two eigenvalue valuations differ, neither x nor such a translate is conjugate into $F^\times K_0$; both orbital integrals vanish. Scalar translations contribute only the specified unramified central-character factor. This proves the ordinary split-torus invariance. For a quadratic field E/F, Proposition 4.2 gives $\ker(P_n : E^\times/F^\times \to E^\times/F^\times) = 1$. The final statement invokes the *published* unit matching, not the elementary split-torus calculation. Its passage to the specified ambient cover is a distinct normalization obligation. □

Remark 8.4 (Scope of the published fundamental lemma)**.** The Flicker–Kazhdan result is a theorem for the matching unramified Hecke units, not an existence result for arbitrarily prescribed functions at finitely many exceptional places. Its original hypotheses include $p \nmid n$ and $(n, N_r) = 1$. In rank two the only maximal-torus types are split and quadratic field tori, so the ordinary invariance calculation above is exhaustive. At higher ranks, mixed tori do not automatically invalidate spherical matching: the Flicker–Kazhdan theorem covers them as well whenever its hypotheses apply.

A regularity and germ obligation. Equation (41) is imposed only where x^n is regular. On a split torus, x can be regular while x^n is singular (Remark 3.3); the power map is a local analytic isomorphism on the torus, but normalized orbital integrals of a function at images approaching a singular element may have nonconstant Shalika-germ behavior. A full transfer construction must either arrange support away from these root-collision loci or prove the matching germ conditions there. The same issue occurs at the complex places, although the cover there splits: the ordinary power map can still carry regular elements to singular ones. No universal vanishing of the germ coefficients is claimed here.

Proposition 8.5 (**[P]** Conditional geometric comparison with an explicit normalization factor)**.** *Fix the rational splittings and compatible central characters of Parts I–III, and fix* corresponding *local Haar measures, global torus measures, and normalizer conventions on the ordinary and covered sides. Choose factorizable ordinary and genuine test functions in a trace-class, regular elliptic support range for which the two simple trace formulas apply. Assume also that* every nongood rational noncentral class has zero ordinary geometric contribution*; for example the test functions may be chosen to suppress the finitely many such classes in a specified simple trace-formula support range. Assume:*

(i) For each $v \notin S$, f_v is FK-compatible as in Definition 8.1, and a genuine test function $\widetilde{f}_v$ satisfies (41) for every regular rational local component s_v of a globally good class $[s^n]$, with a nonzero constant λ_v independent of s_v and of its local torus type. The prescribed central-translate identity holds, $\lambda_v = 1$ for almost all v, and the hypotheses include the complex places.

(ii) For each $v \in S$, use the Part III, Theorem 9.1, orbital-integral transfer $\widetilde{f}_v = \mathcal{O}_{D,v} f_v$, with coefficient $b_{D,v} = |n|_v^{-(r-1)/2}$ and matching quotient measures on H_v/Z_v and $\widetilde{H}_v/\widetilde{Z}_v$. The functions obey the reciprocal central transformation laws.

(iii) The common rational splitting and all local lifts satisfy the star compatibility $\iota(s^n) = \iota(s)^n$; the global torus coefficients (including $|W_s|$), central phases, and quotient measures agree under the stipulated identifications. The support condition removes the central identity contribution. The ordinary and covered simple geometric expansions are absolutely convergent in the selected test-function range.

Put

$$\boxed{\Lambda = \Big(\prod_{v \notin S} \lambda_v\Big)\Big(\prod_{v \in S} b_{D,v}\Big).} \tag{42}$$

Then the regular elliptic geometric distributions, and hence the corresponding simple spectral trace distributions, satisfy

$$\boxed{I_{\widetilde{H}}(\widetilde{f}) = \Lambda\, I_H(f).} \tag{43}$$

In particular the distributions agree if $\Lambda = 1$.

Proof. Since the rational splitting ι is a homomorphism, $\iota(s^n) = \iota(s)^n$; at every completion, $\iota(s)_v$ is a lift of s_v, so this equals its star lift s_v^*. By Theorem 5.2, globally good noncentral rational classes are parametrized by the unique global root class $[s]$ of $[s^n]$, preserving normalizer orders. The explicit nongood-support hypothesis removes the ordinary classes without a covered counterpart; the cover has zero regular orbital integrals on nongood lifts. Local compatibility handles the multiple *local* roots without introducing additional rational summands. Hypothesis (i) supplies $\lambda_v J_{H_v}(s_v, f_v)$ off S, and (ii) supplies $b_{D,v} J_{H_v}(s_v, f_v)$ on S. By (iii) the remaining torus coefficients and central phases agree, and the product of Weyl discriminants is 1 by (36). Every regular rational term on the covered side is thus Λ times the corresponding ordinary term. The central identity has been suppressed, and the stipulated trace-class and convergence conditions permit summation and the two trace formulas. This gives (43). The proof supplies *no* existence theorem for the matching functions, their exceptional local constants, or their necessary nongood support restriction. □

Remark 8.6 (Normalization versus canonical matching)**.** The condition $\Lambda = 1$ is a normalization once nonzero matching functions are known: rescaling one $\widetilde{f}_v$ rescales its λ_v and hence Λ. It is *not* a theorem about canonically normalized units; rescaling a unit changes its canonical value. The meaningful unresolved question is to compute the constants for the prescribed local transfers, especially the finitely many exceptional split places, and to verify the global product in those fixed conventions. Likewise, compactness of $D^\times(K)Z(\mathbb{A}_K)\backslash D^\times(\mathbb{A}_K)$ eliminates continuous spectrum but does not by itself justify every unrestricted trace sum; the support and trace-class conditions above are retained for this reason.

8.4 The identity class and an explicit normalization test

Because r is prime and D is division, every noncentral rational element generates a degree-r field and is regular elliptic. Thus, after quotienting by the rational center, the central identity class is the only remaining singular rational class. Proposition 8.5 suppresses it, but it can be used as an additional test *if* the test range is enlarged and the necessary local germ identities are supplied.

Remark 8.7 (Central discriminants versus orbital-integral germs)**.** Let $d = r(r-1)/2$. For any torus, as a regular $x \to 1$ with x^n regular, the *discriminant ratio* satisfies

$$\frac{\Delta(x)}{\Delta(x^n)} \longrightarrow |n|_v^{-d},$$

because $(1-u^n)/(1-u) \to n$ for each root value $u \to 1$. However this identity *does not by itself yield* $\widetilde{f}_v(1) = \lambda_v |n|_v^{-d} f_v(1)$ from (41). On the split group, normalized orbital integrals generally have nontrivial Shalika germs near singular central elements; their limiting coefficients need not be the evaluation of the test function at 1. The requisite relation between *central test-function values* is therefore a separate local matching obligation, even when the regular orbital integrals match. Compare the unit-specific calculation of Proposition 8.3 with the general germ warning following Lemma 8.2.

At $v \in S$, Part III, Corollary 9.3 gives $\widetilde{f}_v(1) = c_{D,v} f_v(1)$ with $c_{D,v} = b_{D,v} |n|_v^{-r(r-1)/2}$. If, as an *additional unproved local identity-value hypothesis*, every split place (including complex places) satisfies $\widetilde{f}_v(1) = \lambda_v |n|_v^{-d} f_v(1)$ with $d = r(r-1)/2$, and all denominators are nonzero, the ratio of global identity-function values becomes

$$\prod_v \frac{\widetilde{f}_v(1)}{f_v(1)} = \Lambda \prod_v |n|_v^{-r(r-1)/2} = \Lambda \tag{44}$$

by the product formula. This does not imply equality of the *identity trace-formula terms* without a separate choice of matching quotient measures: homeomorphic adelic quotients need not have equal volumes with arbitrarily scaled Haar measures. Nor does it justify retaining the identity term in the simple trace formula without checking its other hypotheses.

Example 8.8 (A possible global numerical ledger)**.** Let $K = \mathbb{Q}(\zeta_3)$, $r = 2$, $n = 3$, and choose a quaternion division K-algebra ramified at the place over 3 and at one other finite place. At the wild division place, $b_{D,v} = 3$, $c_{D,v} = 27$. If the finite split places have matching $\lambda_v = 1$, the equation $\Lambda = 1$ would require $\lambda_\infty = 1/3$ at the unique complex place. Since the normalized complex absolute value has $|3|_\infty = 9$, the hypothesized central-value formula would then give the complex identity factor $(1/3)/9 = 1/27$, balancing 27. This is a *necessary numerical consistency check* under the stated assumptions, not a proof of matching complex test functions or a global comparison.

8.5 Why Li, Kottwitz, and DKV do not close the gap

The linear Jacquet–Langlands theorem of Deligne–Kazhdan–Vignéras [1] controls the ordinary $D^\times$–GL_r relation, including the elliptic character sign, but it does not automatically manufacture a genuine representation of $\overline{G_{D_v}}$. Kottwitz's analysis of stable conjugacy and relative trace coefficients [5] provides the correct arithmetic setting; for the induced tori $\mathrm{Res}_{E/K}\,\mathbb{G}_m$ one has $H^1(K, T) = 1$ by Shapiro's lemma and Hilbert 90, and embeddings $E \hookrightarrow D$ are controlled by Brauer splitting. Wen–Wei Li proves an invariant trace formula for adelic covers under an invariant trace Paley–Wiener hypothesis for Levi subgroups at *real* archimedean places [7]. Here K is totally imaginary, so that real-place hypothesis is vacuous. At the complex places the specified covers split (Part I, Appendix B, item (R4); see also its ambient-cover construction), and the linear invariant Paley–Wiener theorem of Clozel–Delorme [6] supplies the relevant archimedean background. Li also discusses known real metaplectic GL cases, but his general result does not assert the particular *equality* of ordinary-$D^\times$ and covered-$D^\times$ traces required in (35). The simpler already-proved trace formulas of Parts I–II remain the appropriate initial comparison.

9 Spectral separation and a research program

9.1 The actual spectral distribution identity

Let $H = D^\times$ and $\widetilde{H} = \overline{G_D}$ globally, with compatible central characters $\theta, \widetilde{\theta}$. The quotient $H(K)Z_H(\mathbb{A}_K)\backslash H(\mathbb{A}_K)$ is compact because D is a division algebra over K; its covered analogue is compact modulo the specified central preimage as well. Thus the fixed-central-character right regular representations have discrete Hilbert decompositions. For factorizable test functions with the inverse central transformation law, define

$$I_H(f) = \sum_{\rho \in \mathcal{A}(H,\theta)} m_H(\rho) \prod_v \operatorname{tr} \rho_v(f_v), \tag{45}$$

$$I_{\widetilde{H}}(\widetilde{f}) = \sum_{\widetilde{\rho} \in \mathcal{A}(\widetilde{H},\widetilde{\theta})} m_{\widetilde{H}}(\widetilde{\rho}) \prod_v \operatorname{tr} \widetilde{\rho}_v(\widetilde{f}_v). \tag{46}$$

Here $\mathcal{A}$ denotes the relevant automorphic spectrum and m is its actual spectral multiplicity, not an assumed 1. The expressions are interpreted on trace-class test functions satisfying the simple trace-formula hypotheses; a bare unrestricted infinite sum is not justified by notation alone. Under the assumptions of Proposition 8.5, the comparison of geometric expansions supplies $I_{\widetilde{H}}(\widetilde{f}) = \Lambda I_H(f)$; ordinary equality holds when the fixed normalizations give $\Lambda = 1$. It does not identify summands.

This is a conditional identity of *distributions*, with both the multiplicities and local trace factors retained.

9.2 Individual local characters and integrality

An equality of global spectral sums is a statement about *distributions*. To isolate a single local irreducible at a wild place, one needs an argument that the relevant coefficients are integers (or multiplicities which can be shown to isolate a single constituent). Linear independence alone can separate finite lists of characters; it does not supply the required integrality from a unitary Fourier transform.

Remark 9.1 (What an integer spectral coefficient would prove)**.** If a justified global-to-local argument supplies $a_{\rho\tau} \in \mathbb{Z}$ for every τ in the row of a fixed ρ, Proposition 6.1 (Part III, Corollary 5.2) forces $L_{D,v}\Theta_\rho$ to be one irreducible genuine character with positive sign. If integrality holds for all rows, unitarity implies a bijection. This is an application of an already proved lattice criterion, not a second conditional theorem; the missing step is precisely the integrality hypothesis at a wild place.

Separating the remaining tasks. First, the toral Fourier results in Proposition 6.5 establish a necessary integrality property on every quadratic maximal torus in the rank-two wild example, but do not solve the nonabelian gluing problem of Subsection 6.4. The latter requires an actual twisted finite character table and a proof that the chosen finite level captures the desired row. Second, at split places the root sum in Zou's (3.2) should be written against the kernel and index in Proposition 4.2, including the chosen Haar measures and central phases. Third, one needs the explicit ambient-cover identification that transports the known unramified unit matching of [2, §12] to the selected convention, together with functions at the finitely many exceptional split places and a complete global coefficient identity as in Proposition 8.5. The ordinary test functions must also suppress nongood rational classes in the chosen global support range. Finally, one must justify globalization and spectral separation for representations of the covered inner form, and remove the auxiliary supercuspidal and non-cancellation hypotheses of Parts I–II. These are mathematical work packages, not restatements of an already-proved theorem.

10 Selected glossary

Good element A regular lift commuting with the full preimage of its ordinary centralizer; genuine invariant characters vanish on the nongood regular locus.

Scalar-central cover
A cover whose entire inverse image of $F^\times I_r$ is central; here forced by $n \mid r-1+2rc_v$.

Power-map kernel
$\ker(P_n)$ on $T(F)/F^\times$, distinct from the image index. The kernel can be nontrivial on product tori even when the corresponding field-torus map is injective.

Star map $h^* = \widetilde{h}^{\,n}$, independent of lift; multiplication on tori uses n odd.

Weyl discriminant
The factor $\Delta = |\det(1 - \mathrm{Ad}(t)|\mathfrak{g}/\mathfrak{t})|^{1/2}$; it is not the unnormalized characteristic-polynomial discriminant.

Toral Fourier test
: Expand the discriminant ratio R_E in characters of the compact abelian group $E^\times/F^\times$. Integral positive coefficients ensure torus-restriction positivity, not full-group integrality.

Nonabelian gluing
: The additional requirement that compatible genuine covered-torus characters arise from an integral character of the full group $\widetilde{D^\times}$.

Twisted finite quotient
: A finite quotient of H/Z with the central character retained as a multiplier, used to compute projective/genuine irreducible characters.

Packet identity
: An identity of traces over collections of automorphic representations; it does not assert a bijection between individual local irreducibles.

Formal degree
: The Plancherel normalization of a square-integrable representation relative to a specified quotient Haar measure.

Root-sum relation
: Zou's equation (3.2), which sums over all nth roots satisfying the stated regularity condition rather than choosing one arbitrarily.

Acknowledgments

This Part IV manuscript was prepared with mathematical and editorial assistance from ChatGPT (OpenAI). Claude (Anthropic) supplied referee reports on Drafts 1–3, including the correction to the Haar-modulus proof and the split-place matching obligations. Earlier Parts I–III were developed with separate assistance acknowledged there. The author contributions and the referee corrections are documented in the separate revision log. The rank-two Fourier calculations and conditional proof strategy were added in Draft 5 after a separate comparison of Parts I–III and [JC]. The remaining research problems are recorded in the status table and Section 9.

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